Harsh Upadhayay

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Professional Summary

Backend & Cloud Engineer with 3+ years of production experience (2 years full-time) designing distributed systems, event-driven architectures, and AWS cloud-native infrastructure in Python and Go. Owns greenfield architecture from zero to production; delivered 95%+ runtime reductions, cut MTTD from 30 minutes to 1 minute, reduced AWS spend by $1,000+/month, and built fault-tolerant, idempotent backend services using ECS Fargate, Lambda, Kafka/RabbitMQ, Celery, Terraform, and CDK.

Certifications

- **AWS DevOps Professional (86%)**
- **AWS SysOps Administrator Assoc. (88%)**
- **AWS Developer Associate (86%)**
- **AWS Solutions Architect Professional (86%)**
- **AWS Solutions Architect Assoc. (90%)**

Experience

SMS DataTech — **Tokyo, Japan**

Backend & Cloud Engineer (Full-time) — *Oct 2024 – Present*

Software Engineering Intern — *Jul 2023 – Sep 2024*

- **AI Scraping Platform – Greenfield System Design:** Owned end-to-end architecture for a distributed, AI-powered scraping suite from zero to production; designed a fault-tolerant, event-driven orchestration layer (Django/Celery, RabbitMQ) with parallel task batching, automated retries, idempotency, and Dead Letter Queues. Deployed containerized microservices to ECS Fargate via AWS CDK (IaC), enforcing VPC isolation, Secrets Manager, and Fargate Spot; built custom Lambda/EventBridge auto-scaling to eliminate idle compute waste.
- **Cloud Cost Optimization & Infrastructure Ownership:** Reduced total AWS spend by $1,000+/month as co-primary AWS Administrator; cut compute costs $500/month via Savings Plans, Graviton migration, and Spot Instances. Diagnosed a 1.2 TB Aurora RDS bloat, executed a non-blocking batched deletion over 3 days (zero downtime), shrinking the table 83% and cutting RDS costs 50% ($500/month); archived snapshots to Glacier Deep Archive.
- **Enterprise Data Pipeline (Sony Group):** Assumed full ownership of a legacy large-scale collection system across 14 media/e-commerce platforms; refactored 6 major scrapers, migrated normalization logic into Scrapy pipelines, and eliminated intermediary Django/Celery/DB steps – achieving a **95% runtime reduction**. Onboarded and mentored a replacement FTE while maintaining uninterrupted daily/weekly S3 data delivery.
- **PoGo Observability Platform – Event-Driven Monitoring:** Re-architected a polling-based monitor for 20+ servers and 40+ distributed scrapers into an Apache Kafka async pipeline, reducing MTTD from 30 min to 1 min and eliminating all client-reported escaped defects. Built a CPU/RAM/Disk dashboard with 14-day history, cutting per-incident monitoring overhead from 3 hours to 5 minutes.

Technical Skills

- **Languages & Frameworks:** Python, Go, C++, C, SQL, Django, FastAPI, Scrapy, Pandas, NumPy, REST APIs, PostgreSQL, MySQL | **Backend:** Distributed Systems, Microservices, Event-Driven Architecture, Workflow Orchestration, Concurrency, Idempotency, System Design, Scalability, Fault Tolerance, Async Processing (Celery)
- **Cloud & DevOps:** AWS (ECS Fargate, EC2, Lambda, EventBridge, CloudFormation, CDK, CloudFront, RDS/Aurora, S3, IAM, Secrets Manager, CloudWatch), Terraform, Docker, Kafka, RabbitMQ, Redis, CI/CD, Linux, IaC | **Observability:** Monitoring, Logging, Alerting, Incident Response, Production Reliability

Projects

Distributed Rate Limiter (Go) | *Go, Redis, Concurrency, Distributed Systems, System Design*

- Built a distributed rate-limiting library in Go with interface-based algorithm abstraction (Fixed Window, Token Bucket) behind a pure, side-effect-free `Decide(now, state)` contract and a pluggable `StateStore` (`Get` + `CompareAndSwap`) that decouples storage from logic – enabling deterministic testing and hot-swappable in-memory/Redis backends.
- Guaranteed correctness under concurrent mutation via a bounded CAS retry loop that re-evaluates `Decide` on each attempt; eliminated single-mutex key serialization with FNV-hash sharded lock striping to restore parallel throughput (benchmark-verified). Documented with 57 Architecture Decision Records covering core concurrency and design trade-offs.

Self-Hosted Homelab (40+ Services, AI + Cloud Stack) | *Docker, Traefik v3, Authelia, LLDAP, Prometheus, Grafana, Jenkins, Ollama, Cloudflare*

- Architected a domain-driven, containerized infrastructure with Traefik v3 as edge router, Authelia ForwardAuth + OIDC, and LLDAP for identity management across 40+ services; enforced per-service Zero-Trust access policies with secure Cloudflare ingress and segmented network architecture. Integrated full observability (Prometheus, Grafana), CI/CD pipelines (Jenkins + DinD), and GPU-enabled local AI services (Ollama with OpenAI-compatible API) with centralized stateful storage.

Education

Indian Institute of Information Technology, Nagpur — **May 2024**

*Bachelor of Technology in Computer Science and Engineering (**8.5 CGPA**)* — *Nagpur, India*

Kendriya Vidyalaya O.N.G.C. — **March 2019**

*Senior Secondary (**94.4%**), High School (**10 CGPA**)* — *Dehradun, India*

Achievements & Profiles

- Solved **1000+** DSA problems on LeetCode and Codeforces.
- Achieved Specialist rating (**1415 max**) on Codeforces.
- Published **5+** technical articles on GeeksforGeeks.
- Coursera: Supervised ML & Adv. Learning Algorithms.